



Let's Build a Map Quiz with JavaScript!

What is JavaScript?

Every website uses three languages together:

HTML

Structure
(Rangoli outline)

CSS

Style & Colours
(Fills the Rangoli)

JavaScript

Interactivity
(The Diya that lights it up!)

Real-World JavaScript Examples:

- Google search suggestions appear as you type
- Online quizzes check your answers instantly
- Shopping sites update your cart without reloading
- Google Maps moves and zooms when you drag





Variables - The Dabba for Data

A variable is like a dabba (lunchbox) - it stores something inside with a label on top.

```
// Variables store data for our quiz  
  
let correctState = 'Rajasthan';  
let playerScore = 0;  
let message = 'Click a state to answer!';
```

Think of it this way:

Dabba Label = Variable Name (correctState)

Food Inside = Value ('Rajasthan')

You can open the dabba and change the food anytime!

Try it yourself: Change 'Rajasthan' to your home state.
What happens when you change the value inside the dabba?

Functions - School Experiment Steps

A function is like a science experiment procedure - write the steps once, repeat anytime!

```
// A function groups steps together

function checkAnswer(clicked) {
  let correct = 'Rajasthan';
  if (clicked === correct) {
    alert('Correct! Well done!');
  } else {
    alert('Oops! Try again.');
  }
}

// Call the function:
checkAnswer('Rajasthan');
```

Think of it this way:

Experiment Name = Function Name (checkAnswer)

Steps Written on Card = Code inside { }

Saying 'Let's do the experiment!' = Calling the function



Try it yourself: Change the alert messages to something fun!
What happens when you call checkAnswer('Kerala')?



If-Else - The Cricket Umpire

If-else is like a cricket umpire - checks a condition and gives a decision!

```
// The umpire checks: is the answer correct?

function checkAnswer(clicked) {
  let correct = 'Rajasthan';

  if (clicked === correct) {
    alert('Correct! You know your states!');
  } else {
    alert('Wrong! The answer was ' + correct);
  }
}
```

If TRUE (Out!):

```
clicked === correct
alert('Correct!')
```

Else FALSE (Not Out!):

```
clicked !== correct
alert('Wrong!')
```

Try it yourself: What if you check `clicked === 'Kerala'` but click 'Tamil Nadu'?

Click Events - The Doorbell

A click event is like a doorbell - when someone presses it, something happens inside!

```
// HTML button with an onclick event:  
// <button onclick="checkAnswer('Rajasthan')">  
//   Click Rajasthan  
// </button>  
  
// When the button is clicked, checkAnswer runs!  
// The doorbell (click) triggers the action (function)
```

Think of it this way:

Doorbell Button = HTML Button
Pressing the Doorbell = Clicking the Button
Someone Opening the Door = Function Runs

Try it yourself: Add a second button for 'Kerala' that calls checkAnswer('Kerala')



See the Magic! Our Finished Quiz

Here's what we're building today - an interactive India Map Quiz!

Interactive Map Quiz Demo

Click Any State to Play!

How it works:

1. A map of India is shown with clickable states
2. The quiz asks: 'Which state is Rajasthan?'
3. You click a state - JavaScript checks your answer
4. You get instant feedback: Correct or Try Again!



Map Quiz Core - Steps 1 & 2

Let's build our quiz step by step - combining all four concepts!



Step 1: Variable

```
let correctState = 'Rajasthan';
```

Step 2: Function

```
function checkAnswer(clicked) {  
  // All our quiz logic goes inside here  
  // This function runs when a state is clicked  
}
```

Variable = stores the correct answer | Function = groups all the checking steps

Map Quiz Core - Steps 3 & 4

Now we add the decision-making and the trigger!



Step 3: If-Else

```
function checkAnswer(clicked) {  
  let correct = 'Rajasthan';  
  if (clicked === correct) {  
    alert('Correct! Well done!');  
  } else {  
    alert('Oops! The answer was ' + correct);  
  }  
}
```

Step 4: Click Event (HTML)

```
<button onclick="checkAnswer('Rajasthan')">  
  Click Rajasthan  
</button>  
  
<button onclick="checkAnswer('Kerala')">  
  Click Kerala  
</button>
```

If-Else = makes the decision | Click Event = triggers the function

The Complete Quiz Code

All four building blocks combined into one working quiz!

```
// Step 1: Variable
let correctState = 'Rajasthan';

// Step 2 & 3: Function with If-Else
function checkAnswer(clicked) {
  if (clicked === correctState) {
    alert('Correct! You know your states!');
  } else {
    alert('Wrong! The answer was ' + correctState);
  }
}

// Step 4: HTML Click Event
// <button onclick="checkAnswer('Rajasthan')">Rajasthan</button>
// <button onclick="checkAnswer('Kerala')">Kerala</button>
```



Correct Answer Trace:

Click 'Rajasthan'
clicked === correctState = TRUE
alert('Correct! You know your states!')

Wrong Answer Trace:

Click 'Kerala'
clicked === correctState = FALSE
alert('Wrong! The answer was Rajasthan')



Your Turn! Challenges

Try these coding challenges to level up your quiz!

Level 1: Change the Question

Change correctState to 'Kerala' and update the button text to match.

```
let correctState = 'Kerala';
```

Level 2: Better Messages

Make the alert say which state was clicked:

```
alert('You clicked ' + clicked + ' but the answer is ' + correctState);
```

Level 3: Add a Second Question

Create a new variable and function for a second question:

```
let correctCapital = 'Jaipur';
```

Level 4: Display Score

Keep score and show it on the page:

```
let score = 0;
```

```
document.getElementById('score').innerText = score;
```

Quick Reference Card

Concept	Analogy	Code Syntax
Variable	Dabba (Tiffin Box)	<code>let correctState = 'Rajasthan';</code>
Function	Experiment Steps	<code>function checkAnswer(clicked) { }</code>
If-Else	Cricket Umpire	<code>if (clicked === correct) { } else { }</code>
Click Event	Doorbell	<code>onclick="checkAnswer('Rajasthan')"</code>
Alert	Loudspeaker	<code>alert('Correct! Well done!');</code>

Full Quiz Skeleton Code:

```
let correctState = 'Rajasthan';

function checkAnswer(clicked) {
  if (clicked === correctState) {
    alert('Correct! You know your states!');
  } else {
    alert('Wrong! The answer was ' + correctState);
  }
}

// HTML:
// <button onclick="checkAnswer('Rajasthan')">Rajasthan</button>
// <button onclick="checkAnswer('Kerala')">Kerala</button>
```



Exit Ticket

Before you go, answer this question to show what you learned today!

What is an interactive webpage?

Name one thing JavaScript can do that HTML and CSS cannot.

More to think about:

1. What does a variable do? (Think: Dabba)
2. Why do we use functions? (Think: Experiment)
3. How does if-else help our quiz? (Think: Umpire)
4. What triggers our checkAnswer function? (Think: Doorbell)



Thank You & Keep Creating!

You built your first interactive quiz with JavaScript!

Remember:

Variables store data | Functions group steps
If-Else makes decisions | Click Events trigger actions

